Royal Devon University Healthcare NHS Foundation Trust

Patient Mobilisation with Continuous Renal Replacement Therapy Standard Operating Procedure Trustwide	
Post holder responsible for Procedural Document	[1]Advanced Respiratory Physiotherapist for ICU [2]Consultant in Intensive Care Medicine [3] Senior Sister for ICU
Author of Standard Operating Procedure	Ruth Wood[1] Nicholas Love[2]Judith Bassett[3]
Division/ Department responsible for Procedural Document	Clinical Support & Specialist Services
Contact details	Ruthwood4@nhs.net
Date of original standard operating procedure	03/11/2022
Impact Assessment performed	No
Approving body and date approved	Clinical Effectiveness Committee: 10/12/2025
Governance approvals	ICU SGG East 01.04.2025 ICU SGG North 28.02.2025 Therapies Guideline Group 20.05.2025 Clinical Specialist Services Care Group Governance 27.08.2025 Surgery Care Group Governance 13.11.2025
Review date	30.12.2027
Expiry date	21.06.2028
Date document becomes live	15.01.2026

Please *specify* standard/criterion numbers and tick ✓ other boxes as appropriate

Monitoring Information		Strategic Directions – Key Milestones	
Patient Experience		Maintain Operational Service Delivery	
Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:	GPICS 2022		
Note: This document has been assessed for any equality, diversity or human rights implications			

Controlled document

This document has been created following the Royal Devon University Healthcare NHS Foundation Trust. It should not be altered in any way without the express permission of the author or their representative.

Full History		Status: Final	
Version	Date	Author (Title not name)	Reason
1.0	04/11/2022	Ruth Wood Nicholas Love Judith Bassett	New document to meet GPICS standards for 2022.
		Consultation with relevant groups and governance groups	Band 6/7/8 respiratory physiotherapists Consultation April / May 2023 ICU specialist nurse band 7 2/5/23 ICU senior team Governance meeting 26/5/23 AHP team leaders 13/6/23 Shared with Exeter ICU physiotherapy lead 20/06/23
1.1	20/2/2025	Ruth Wood	Completed literature review and 9 studies read. More evidence supporting. Consultation with eastern colleagues in ICU and Therapies to enable this document to be Trustwide
2.0	10/12/2025	Ruth Wood	Final Version presented to Clinical Effectiveness Committee

Associated Trust Policies/ Procedural documents:	N/A
Key Words:	Intensive care unit, Early mobility Mobility ICU Critical care Renal replacement therapy CRRT RTT CVVH Continuous Veno Venous Haemofiltration
In consultation with and date: Band 6 / 7 / 8 Respiratory Physiotherapists April/ May 2023 ICU senior team and ICU governance 26/5/23 AHP team leaders / 13/6/23 Shared with Exeter ICU physiotherapy lead 20/06/23	
Contact for Review:	Ruth Wood

CONTENTS

KEY POINTS OF THIS PROCEDURAL DOCUMENT: 4
1. INTRODUCTION AND SUPPORTING INFORMATION 5
2. PURPOSE 5
3. DEFINITIONS 5
4. DUTIES AND RESPONSIBILITIES OF STAFF 5
5. MAIN BODY OF STANDARD OPERATING PROCEDURE 5
6. KEY MESSAGES 7
7. WHO IS THIS DOCUMENT FOR? 7
8. WHEN SHOULD THIS DOCUMENT BE USED? 7
9. CONSIDERATIONS 7
10. PROCEDURAL STEPS 7
11. ARCHIVING ARRANGEMENTS 10
13. REFERENCES 10
APPENDIX 1: COMMUNICATION PLAN 12

KEY POINTS OF THIS PROCEDURAL DOCUMENT:

This document outlines the safe mobilisation of CRRT in Intensive Care Units.

1. INTRODUCTION AND SUPPORTING INFORMATION

1.1 Timely early rehabilitation is well documented to be safe and expected in the Intensive Care Unit (ICU). More recent evidence suggests that early mobilisation can improve duration of the filter set.

1.2 Most patients fall into two categories while being treated for Acute Kidney Injury (AKI).

- patients that are at their baseline mobility (normally independent) but require single organ support in the form of Continuous Renal Replacement Therapy (CRRT).

- patients that require multiorgan support and are way off their baseline mobility due to prolonged ventilation. These patients are ventilated and are having ongoing physiotherapy as part of regular rehabilitation but require CRRT as a necessary addition to the patient's current care on the ICU.

Both situations pose challenges and therefore require careful consideration as a multidisciplinary team (MDT) and each patient should be considered individually.

1.3 This SOP has been developed to define the best practice for rehabilitation and mobilisation of adult patients on filtration and to give guidance on how this can be achieved safely and effectively.

2. PURPOSE

2.1 Timely to define best practice in the mobilisation of patients requiring Continuous Renal Replacement Therapy (CRRT) to ensure safe practice.

3. DEFINITIONS

ICU - Intensive Care Unit
AKI - Acute Kidney Injury
CRRT - Continuous Renal Replacement Therapy
MDT - Multidisciplinary team

4. DUTIES AND RESPONSIBILITIES OF STAFF

The Title of Post holder is responsible for dissemination of the policy.

5. MAIN BODY OF STANDARD OPERATING PROCEDURE

5.1 **Patients that are at their baseline mobility (normally independent) but require single organ support in the form of CRRT**

If the patient is at baseline mobility and keen to mobilise away from the bed space a discussion between the consultant and nurses involved in care (bed space and nurse in charge) should occur. Considerations of putting the Continuous Veno Venous Hemofiltration (CVVH) into treatment break mode for the time of the mobility and how

long this break can occur for should be agreed. If you click on treatment break the machine gives you instructions to follow. This break can last for 4 hours. Essentially, for a 4 hour break the blood is flushed back using a 500ml bag of 0.9% saline. The access line and return line are connected to the saline bag for the machine to carry on in recirculation. The vascath is flushed with 0.9% saline and heparin. The patient can then do the move/walk and then reattach.

5.2 **Patients that require multiorgan support and are off their baseline mobility:**

These patients are ventilated via a tracheostomy or endotracheal tube and are having ongoing physiotherapy as part of regular rehabilitation but require CRRT as a necessary addition to their current care on the ICU.

This type of patient should be assessed by a physiotherapist that has adequate experience in mobilising a patient receiving CRRT. This should be a discussion between the physiotherapist and the band 7 physiotherapy advanced practitioner within supervision/competency review. This is most likely to be a band 6 respiratory physiotherapist or above.

Consideration to mobilise a patient on the filter should be discussed with the consultant to see if it is appropriate and this can then be agreed with the nurse in the bed space / in charge. All professionals are encouraged to speak up if they do not feel it is appropriate to mobilise a patient.

It is the joint responsibility of all members of the team to ensure adequate numbers of participating staff / skill mix and agree a suitable time for the mobilisation to occur. It is important to know that the consultant is not busy with a sick patient or in a different building to ensure that they are available if there was an adverse event during the treatment session.

An agreement must be reached on the proposed mobilisation plan between the nurse, consultant, and physiotherapist and be communicated to all staff involved in the manoeuvre. This should include an appropriate action plan should filter complications arise.

The senior physiotherapist will allocate roles to staff assisting with the session and explain what those roles involve ensuring that the aim of the rehab session is clear and the practical plan is explained. Prior to treatment the MDT should discuss any alterations which can be made to the filter during treatment to assist with rehab e.g. flow rate reduction, optimal vas-cath position. There should be discussion between the team on whether it is appropriate to put the filter into ‘care mode’ prior to mobilising. An appropriately experienced member of the team should be charged with monitoring the filter and maintaining vas-cath access throughout treatment and this should be specified when allocating roles. Should the filter alarm for any number of reasons or become compromised at any stage during treatment, appropriate action should be taken to ensure the safety of the patient and adequate running of the filter.

If the patient is sat out at the end of the treatment, a time should be arranged for getting the patient back to bed or a bleep number provided to ensure help to get back to bed. In the first instance if a time has not been arranged or the patient needs to get back in sooner than originally thought then the physiotherapy bleep number is 089 and 327.

6. KEY MESSAGES

- Safe, standardised working practice is essential for the mobilisation of patients on CRRT.
- Rehabilitation is initiated at the earliest, safe, and appropriate time for individual patients.
- Not all patients will be appropriate for mobilisation on CRRT and therefore multidisciplinary discussion and agreement is essential.
- Inadequate resources may on occasion prevent mobilisation on safety ground.
- Appropriate patients are selected with clear communication for this type of rehabilitation with multi-disciplinary agreement.
- Clear communication and teamwork are utilised to minimise risk of harm.
- All ICU staff should be aware of this

7. WHO IS THIS DOCUMENT FOR?

- ICU nurses.
- Consultants.
- Physiotherapists.
- Health Care Assistants or therapy assistants assisting with the mobilisation of these patients.
- Band 4 nursing associates or trainee nursing associates.
- Student physiotherapists, medical students and student nurses.
- All staff who may have input / give advice towards the rehabilitation.

8. WHEN SHOULD THIS DOCUMENT BE USED?

This SOP must be followed for all patients being considered for mobilisation of patients on CRRT across RDUH.

9. CONSIDERATIONS

In addition to the normal considerations for mobilising a patient on ICU, extra precaution should be taken, and it is appropriate to seek advice from the ICU consultant regarding a safe level of inotropic support while mobilising these patients while on the filter.

10. PROCEDURAL STEPS

Task	Procedure
Preparation for all tasks	• Identify appropriate patient. • Check for suitability (discuss with consultants and nurse in charge as a team).

	• Establish agreement between all relevant members of the MDT including ICU consultant, nurse in bed space / nurse in charge, senior physiotherapist and participating staff. • Document Risk Assessment taking in to account specific risks related to the filter; removal of vas-cath, clotting / clogging of vas-cath, bleeding, unstable BP, pain. • Establish minimum staffing numbers / skill mix for the procedure. Ensure an appropriately qualified staff member is present throughout treatment to monitor and adjust the filter. Ideally this would be patient's designated staff nurse for the day. • If the proposed plan includes mobilisation to a chair, plan the entire task including transfer out of bed, time spent in the chair and method of return to bed. • Coordinate appropriate timing of procedure for both patient and multidisciplinary team, particularly ensuring the consultant is available if an endotracheal tube or tracheostomy is in situ and/or the patient receives inotropic support. Consider specifically if inotropic level appropriate. • Communicate and document the plan to all participating members of staff and the patient.
Transferring patient to edge of bed	• Allocate roles to all staff involved. Ensure all participating staff are aware of the plan and movements involved in the task. • Prepare the patient and environment for the task e.g. removing / disconnecting any unnecessary monitoring or lines. • Designate one person of appropriate skill level to monitor patient's filter and vas-cath. Designate other roles appropriately as per the therapist's discretion. • Transfer patient to edge-of-bed in manner most appropriate to patient's needs and position of line (e.g. through side lying, swivel, hoist etc.) • Appropriate staff member to ensure filter remains secure and running adequately during edge-of-bed assessment. Nursing staff to advise the rest of the team of any filter difficulties and make adjustment to the filter as deemed they see fit. • Physiotherapist conducts full edge-of-bed sitting assessment / rehabilitation session as appropriate.

	• If transferring to a plinth or chair the filter must be able to be delivered during this intervention i.e. appropriate length of line and appropriate vas-cath positioning. • If the patient is currently being filtered via a femoral line, consideration should be made to prevent hip flexion past 90 degrees e.g. perch sitting, sling hoisting with hips in increased extension. • Cardiovascular monitoring must be continuous throughout all treatment and treatment should be adjusted to maintain values within individually defined parameters.
Transferring to chair	• One staff member with appropriate skill will be allocated to maintain security of the vas-cath. An additional member of staff should be allocated to ensure the safety of other lines if required. Minimum staffing requirement for the transfer of the patient should be determined with the exclusion of this staff member. • Transferring to chair may be undertaken as appropriate to the patient's needs. The following considerations should be taken into account. • Sitting out with a femoral vas-cath in situ should only be considered if a chair which can be reclined is available or if legs held at 90 degrees for 1 minute with no issue. Research suggests this may help to ensure there will be adequate flow through the line. • If problems with filter during edge sit or plinth sit arise consider if any adjustments can be made with nursing staff / consultant or if manoeuvre should be discontinued. • Once safely positioned in the chair, the vas-cath should be rechecked that it is running correctly by the staff nurse. • When advising nursing staff on returning patient to bed, identify that extra resource will need to be available to ensure security of the vas-cath on transfer back. Where able, therapy staff should volunteer this resource if required, to facilitate a safe transfer back to bed. Time arranged. Bleep numbers as above.
Walking	• Mobilising on the spot is feasible and often easier due to length of line and power cord. This is a lot easier than mobilising away from the bed space due to the size of the machine and to minimise risk of disruption of the line / flow.

	• If the patient is very mobile / single organ failure it may be worth putting the CRRT in treatment break mode for the time of the mobility. The blood can be flushed back and set recirculated. The line can then be disconnected from the patient, the patient can do the move / walk and then the line can be reattached. This break can last for 4 hours or without return blood can be detached for 15 minutes.

11. ARCHIVING ARRANGEMENTS

The original of this SOP will remain with the author Ruth Wood. An electronic copy will be maintained on the Trust intranet. Archived electronic copies will be stored on the Trust's “archived policies” shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 10 years.

12. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE STANDARD OPERATING PROCEDURE

To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Incidents related to early mobility and CRRT.	Notes audit, attendance at Governance meetings.	ICU Governance Team.

13. REFERENCES

Bento, H., Dummer, D., and Lohse, B (2020) Walking while dialyzing: a retrospective observation of early mobility and ambulation for patients on continuous renal replacement therapy. Critical care explorations, 2.

Crowe, S., Brook, A., and Haljan, G, (2019) Continuous renal replacement therapy and mobilization: yes, it is possible. The Canadian journal of critical care nursing. 30(1) 12-16
Jacobs, F (2015) Early mobilisation on continuous renal replacement therapy is safe and may improve filter life. Critical care 19: 205

Crowe, S., Brook, A., and Reynolds, J (2017) You want me to what? Mobility and Continuous renal replacement therapy (CRRT) patients. 28(2)

Johnson, J., Lohse, B., and Bento, H (2019) Improving outcomes for critically ill cardiovascular patients through increased physical therapy staffing. Arch phys med rehabilitation 100(2) 270-277

Lee, H., Jun Ko, Y., and Jung, Jinhee (2016) Monitoring of potential safety events and vital signs during active mobilization of patients undergoing continuous renal replacement therapy in medical intensive care unit. Blood purification 42:83-90

Mayer, K., Hornsby, A., Soriano, V (2019) Safety, feasibility, and efficiency of early rehabilitation in patients requiring continuous renal replacement: a quality improvement study. 5: 39-47

Raurell- Torreda, M., Regaira- Martinez, E., and Planas- Pascual, B (2021) Early mobilisation algorithm for the critical care patient. Expert recommendations. Enfermeria Intensiva 32: 153-163

Talley, C., Wonnacott, R and Schuette, J (2013) Extending the benefits of Early mobility to Critically ill patients undergoing continuous renal replacement therapy. Critical Care Nursing 36 (1) 89-100

Toonstra, A., Zanni, J., and Sperati, J, Feasibility and safety of physical therapy during continuous renal replacement therapy in the intensive care unit. Thoracic society 13 (5) 699-704

Trumble, D., DeVries, J., and Reil, E (2022) Optimizing MObility for critically ill patiEnts undergoing Continuous Renal Replacement Therapy (MOvE CRRT): An audit of mobility interventions in the intensive care unit, Canadian Journal of Respiratory, Critical Care, and Sleep Medicine, 6(3) 184-192

APPENDIX 1: COMMUNICATION PLAN

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the SOP	ICU nurses, Consultants, Physiotherapists, Health Care Assistants or therapy assistants assisting with the mobilisation of these patients, Band 4 nursing associates or trainee nursing associates. Student physiotherapists, medical students and student nurses. All staff who may have input / give advice towards the rehabilitation.
The key changes if a revised document	For both sites now.
The key objectives	Safety related to early mobility in CRRT Ensure GPICS recommendations are adhered too.
How new staff will be made aware of the procedure and manager action	Physiotherapy: • Induction for ICU / on-call • On-call rotation • On-call document Nursing: • Induction for ICU • Step competency's
Specific Issues to be raised with staff	Safety section
Training available to staff	Induction for ICU / on-call and time spent on ICU with clinicians. On-call mentor in which staff can highlight training needs and feeds into competency documents.
Any other requirements	N/A
Issues following Equality Impact Assessment (if any)	N/A
Location of hard / electronic copy of the document etc.	G:ICU/SOPSICU